

MODULE 04 · DOMAIN 2 - DESIGN RESILIENT ARCHITECTURES

Decoupling and Event-Driven Design

Queues, topics, event buses and workflows.

DOMAIN 2 of 4 EXAM WEIGHT 26% MODULE 4 of 22

A queue turns a **failure** into a **delay**. That is the entire reason the exam loves them.

SQS or SNS?

AMAZON SQS

A queue. One message, one consumer

- Consumers pull when they are ready
- Messages persist until deleted or expired
- Buffers load and smooths spikes
- Retention up to fourteen days

AMAZON SNS

A topic. One message, many subscribers

- Push to every subscriber immediately
- No storage. Undelivered means lost
- Fan-out to many independent systems
- Subscribers include SQS, Lambda, HTTPS, email

THE TIEBREAKER

One message processed once means SQS. The same message needed by several systems means SNS.

The fan-out pattern

SNS and SQS together, and why the exam prefers it to SNS alone.

- ◆ **Publish once to an SNS topic**, with several SQS queues subscribed to it.
- ◆ **Each downstream system gets its own queue**, so one slow consumer does not affect the others.
- ◆ **Messages persist** in each queue, so a consumer that is down loses nothing.
- ◆ **Adding a new consumer** means adding a queue and a subscription. No producer change.
- ◆ **SNS straight to Lambda** is fine for simple cases but has no buffer if Lambda is throttled.

ON THE EXAM

'Several teams need the same event, and none may lose it' is fan-out, not SNS alone.

SQS details the exam asks about

FEATURE	WHAT TO KNOW
Standard queue	Nearly unlimited throughput, at-least-once, best-effort ordering
FIFO queue	Strict ordering and exactly-once, lower throughput. Name must end in .fifo
Visibility timeout	How long a message is hidden after being read. Too short causes double processing
Dead-letter queue	Where messages go after N failed processing attempts. Enables debugging
Long polling	Waits for messages rather than returning empty. Reduces cost and empty responses
Delay queue	Postpones delivery of new messages by up to fifteen minutes
Message size	Up to 256 KB. Larger payloads go in S3 with a pointer in the message

ON THE EXAM

'Messages are being processed twice' is a visibility timeout question. 'Order matters' is FIFO.

EventBridge or SNS?

AMAZON EVENTBRIDGE

Routing on content

- Filter on any field in the event body
- Receives events from AWS services directly
- Schema registry and event archive with replay
- Scheduled rules replace cron

AMAZON SNS

Broadcasting to subscribers

- Very high throughput, very low latency
- Simple filtering on message attributes
- SMS, email and mobile push endpoints
- Cheaper at large volumes

THE TIEBREAKER

Reacting to an AWS service event, or routing by content, means EventBridge.
Broadcasting your own message to many subscribers means SNS.

Orchestration and the rest

- ◆ **AWS Step Functions.** Coordinates multi-step workflows with retries, branching and error handling, without you writing that logic.
- ◆ **Amazon MQ.** Managed ActiveMQ or RabbitMQ. Only correct when migrating an application that already speaks those protocols.
- ◆ **Kinesis Data Streams.** Ordered, replayable stream for many consumers reading the same records. Not a work queue.
- ◆ **API Gateway with SQS integration.** Accepts requests and queues them without any compute in between.

ON THE EXAM

Existing JMS or AMQP application being migrated means Amazon MQ. Anything new means SQS or SNS.

An online ticketing platform releases tickets for major events at advertised times. In the first two minutes of a release, traffic rises to roughly forty times normal levels.

The order processing service writes to a relational database and can handle about three hundred orders per second. During releases it becomes overwhelmed and customers see errors, although the business is content for confirmation emails to arrive minutes later.

Which change best addresses this?

- ☐ A Place orders on an SQS queue and have the processing service consume from it
- ☐ B Increase the RDS instance size to handle forty times the load
- ☐ C Add an Auto Scaling group to the order processing tier
- ☐ D Add an SNS topic that the order service publishes to

◆ PAUSE AND DECIDE

Which change best addresses this?

- A

 SQS queue in front of processing
Orders are accepted instantly and drain at the rate the database can sustain. The spike becomes a delay, which the business has accepted.
- B

 Scale up RDS forty times
Pays for peak capacity every hour of the year to serve two minutes of it, and there is a ceiling on how far a single instance scales.
- C

 Auto Scaling the processing tier
More consumers still hit the same database limit, and scaling takes minutes while the spike lasts two.
- D

 SNS topic
SNS pushes immediately with no storage, so it forwards the spike to the same overwhelmed service rather than absorbing it.

THE SIGNAL

A short, extreme spike plus tolerance for delayed completion is the textbook queue scenario.

When a customer uploads a photograph, four things must happen: a thumbnail is generated, the image is scanned for prohibited content, a search index is updated, and the analytics team receives a record.

These are owned by four separate teams. The analytics consumer is frequently offline for maintenance, and no event may be lost while it is down. New consumers are expected next year.

Which architecture meets these requirements?

- ☐ A The upload service calls each of the four services directly in sequence
- ☐ B Publish to an SNS topic with the four services subscribed directly
- ☐ C Publish to an SNS topic with four SQS queues subscribed, one per consumer
- ☐ D Write to a single SQS queue that all four services read from

◆ PAUSE AND DECIDE

Which architecture meets these requirements?

- A

 Direct sequential calls
Tightly coupled. One slow or offline service blocks the upload, and every new consumer means changing the upload service.
- B

 SNS with direct subscriptions
Close, but SNS does not store. Events published while analytics is offline are lost, which the requirement forbids.
- C

 SNS fan-out to four SQS queues
Each consumer has its own durable buffer, offline consumers lose nothing, and a new consumer is just another subscription.
- D

 One shared SQS queue
A message is consumed once, so whichever service reads it first prevents the other three from seeing it.

THE SIGNAL

The phrase 'no event may be lost while a consumer is down' is what upgrades SNS alone to SNS plus SQS.

A payment reconciliation system reads messages from an SQS standard queue. Operations report that some payments are being recorded twice, and that a small number of malformed messages are being retried endlessly, delaying valid work.

Processing each message takes about forty seconds. The visibility timeout is set to thirty seconds.

Which combination of changes addresses both problems?

- ☐ A Increase the visibility timeout above the processing time, and configure a dead-letter queue
- ☐ B Switch to a FIFO queue and increase the retention period
- ☐ C Enable long polling and reduce the batch size
- ☐ D Add more consumers so that the backlog clears faster

◆ PAUSE AND DECIDE

Which combination of changes addresses both problems?

- A

Raise visibility timeout, add a DLQ
The timeout expiring mid-processing causes the duplicates; the DLQ removes poison messages after a set number of attempts.
- B

FIFO queue and longer retention
FIFO would mask duplicates at high cost to throughput, but the real fault is the timeout. Retention is unrelated to either problem.
- C

Long polling, smaller batches
Both reduce cost and empty receives. Neither affects duplicate processing or endless retries.
- D

More consumers
More consumers processing messages that reappear mid-flight increases duplication rather than reducing it.

THE SIGNAL

Forty seconds of processing under a thirty second timeout is the arithmetic that gives the answer away.

A media company processes video uploads through eight sequential steps, including transcoding, subtitling and publishing. Some steps take hours. If any step fails it must be retried three times, and if it still fails a human must be notified and the job paused for inspection.

The logic currently lives in a long-running Python script on an EC2 instance, and when the instance restarts, in-flight jobs are lost.

Which service should coordinate this workflow?

- ☐ A AWS Step Functions
- ☐ B An SQS queue between each pair of steps
- ☐ C Amazon EventBridge scheduled rules
- ☐ D A larger EC2 instance with the script running as a systemd service

◆ PAUSE AND DECIDE

Which service should coordinate this workflow?

A Step Functions

Built for exactly this: ordered steps, declarative retries, error branches, human approval, and durable state that survives restarts.

C EventBridge scheduled rules

Schedules things at times. It does not track the state of an individual job through a sequence of steps.

B Queue between each step

Would decouple the steps, but you would still write all retry counting, error branching and state tracking yourself.

D Bigger instance

Addresses nothing. The problem is lost state on restart, and a larger instance restarts just as destructively.

THE SIGNAL

Ordered steps, retry counts, error branches and durable state together mean Step Functions. A queue decouples; it does not orchestrate.

A security team wants an automated response whenever an EC2 instance is stopped in any of twelve accounts. The response differs depending on which environment tag the instance carries: production instances page an on-call engineer, while development instances are simply logged.

The team does not want to poll for changes or run any always-on infrastructure.

Which solution meets these requirements?

- ☐ A An EventBridge rule matching the instance state-change event, routing on the tag to different targets
- ☐ B An SNS topic that all twelve accounts publish to when instances stop
- ☐ C A Lambda function on a one-minute schedule that lists instance states
- ☐ D CloudTrail with a CloudWatch Logs metric filter for the StopInstances API call

◆ PAUSE AND DECIDE

Which solution meets these requirements?

- A

EventBridge rule with content routing
EC2 emits state-change events natively, rules can match on event content including tags, and different matches route to different targets.
- B

SNS topic
Something must detect the stop and publish. SNS is the delivery mechanism, not the detector, and it cannot route on event content.
- C

Scheduled Lambda polling
Explicitly excluded by 'does not want to poll', and it would miss instances stopped and started between runs.
- D

CloudTrail metric filter
Can alert that the API was called, but a metric filter cannot branch the response based on a resource tag.

THE SIGNAL

'React to something happening inside AWS' plus 'route based on content' is EventBridge. SNS delivers; it does not detect.

A company is migrating a fifteen-year-old Java order system to AWS. It uses the JMS API to talk to a self-managed ActiveMQ broker, with topics and queues configured in ways the team no longer fully understands.

The migration must complete in four months, and the business will not fund application rewrites during the move.

Which messaging service should be used?

- ☐ A Amazon SQS, converting the JMS calls to the SQS SDK
- ☐ B Amazon MQ running ActiveMQ
- ☐ C Amazon Kinesis Data Streams
- ☐ D Amazon EventBridge with a custom event bus

◆ PAUSE AND DECIDE

Which messaging service should be used?

A SQS with rewritten calls

The better long-term destination, but it requires the application rewrite the business has explicitly refused to fund.

C Kinesis Data Streams

A streaming service for ordered replayable records. It does not speak JMS and is not a broker replacement.

B Amazon MQ with ActiveMQ

Speaks JMS natively, so the application connects with a configuration change rather than a rewrite. Managed by AWS.

D EventBridge custom bus

A routing service with no JMS support. Would require the same rewrite as option A, plus a redesign.

THE SIGNAL

Amazon MQ has one exam signal: an existing application using standard protocols that must not be rewritten.

Module 4 takeaways

- ◆ **A queue turns failure into delay.** Spikes plus tolerance for later completion means SQS.
- ◆ **Several consumers, none may miss an event,** means SNS fan-out to SQS queues.
- ◆ **Duplicate processing** is almost always the visibility timeout being shorter than processing time.
- ◆ **Workflow shape** means Step Functions. Message passing means a queue.
- ◆ **Existing JMS application, no rewrite** means Amazon MQ. Otherwise SQS.

ON THE EXAM

If a question describes one component failing and taking others with it, the answer inserts something between them.